Fiscal Year:	FY 2024	**Task Last Updated:**	FY 11/07/2025
PI Name:	Shelhamer, Mark Sc.D.		
Project Title:	Otolith and Posture Evaluation II		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Baltimore	**State:**	MD
Zip Code:	21205-1832	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	11/01/2021	**End Date:**	10/31/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2025 per TRISH (Ed., 8/26/25). NOTE: End date changed to 03/31/2028 per TRISH (Ed., 3/4/25). NOTE: End date changed to 03/31/2025 per TRISH (Ed., 11/4/24). NOTE: End date changed to 10/31/2024 per TRISH (Ed., 11/4/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sarma, Mallika Ph.D. (Johns Hopkins University) Schubert, Michael Ph.D. (Johns Hopkins Medical School)		
Grant/Contract No.:	NNX16AO69A-EXP0008		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EXPAND Project This project will provide information on binocular alignment as a measure of otolith asymmetry – more specifically as a measure of central neural compensation for this asymmetry. This low-level sensorimotor response has several attractive features: • It is easily and rapidly measured with minimal equipment. • It provides complementary information to posture and locomotion tests. • It has been validated in studies in vestibular patients and in parabolic flight. • It can provide an alternate means to assess underlying neurovestibular function which contributes to many sensorimotor responses. We propose here the use of a simple perceptual test to assess binocular alignment, known as vertical alignment nulling and torsional alignment nulling (VAN/TAN), and associated tests of posture and balance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Motion sickness immediately on return to Earth remains a problem, with 100% of interviewed astronauts reporting symptoms. This project revisits and updates extensive and promising previous work (mostly by Markham and Diamond) on prediction of space motion sickness (SMS) susceptibility. Prediction of SMS (and postflight postural instability) would be of great value in setting crew expectations, suggesting crew roles, and eventually for testing and individualized targeting of countermeasures. In addition, for commercial spaceflight, the ability to mitigate SMS would greatly help this industry in expanding its customer base. A version of the apparatus and procedures developed and refined for this project is in use in the vestibular clinic at Johns Hopkins (co-investigator Michael Schubert) and is being further developed for assessment of military personnel with traumatic injuries.
Task Progress:	This project provides information on binocular alignment as a measure of otolith asymmetry – more specifically, as a measure of central neural compensation for this asymmetry. This low-level sensorimotor response has several attractive features: • It is easily and rapidly measured with minimal equipment. • It provides complementary information to posture and locomotion tests. • It has been validated in studies in vestibular patients and in parabolic flight. • It can provide an alternate means to assess underlying neurovestibular function which contributes to many sensorimotor responses. Results from I4, three Axiom flights, and Polaris Dawn are promising. There is evidence that otolith asymmetry (uncompensated for in space) is associated with SMS incidence, and that the temporal structure of postflight body sway is related to this asymmetry as well. These findings have operational implications (for commercial and government spaceflight), and establish a foundation for refinement of the procedures to improve predictive capability and for expansion to include multisensory integration. The finding of possible effects of otolith asymmetry on posture is novel and warrants study on its own merits. Findings are hampered by a lack of consistency in acquiring flight-crew data, leading to problems with data integrity. The protocol changed from flight to flight, addition of HRP measures caused confusion, and access to SMS information has been sporadic. This is easily fixable by regular review and reevaluation of procedures, examination of data as it becomes available (for quality control), and a commitment to obtaining consistent data across as many crewmembers and test sessions as possible.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Jones CW, Overbey EG, Lacombe J, Ecker AJ, Meydan C, Ryon K, Tierney B, Damle N, MacKay M, Afshin EE, Foox J, Park J, Nelson T, Suhail M, Byhaqui SG, Aslam B, Tali UA, Nisa L, Menon P, Patel CO, Khan SA, Ebert DJ, Everson A, Schubert MC, Ali NN, Sarma MS, Kim J, Houerbi N, Grigorev K, Garcia Medina S, Summers AJ, Gu J, Altin JA, Fattahi A, Hirzallah MI, Wu JH, Stahn AC, Beheshti A, Klotz R, Ortiz V, Yu M, Patras L, Matei I, Lyden D, Melnick A, Banerjee N, Mullane S, Kleinman A, Loesche M, Menon AS, Donoviel DB, Urquieta E, Mateus J, Sargsyan AE, Shelhamer M, Zenhausern F, Bershad EM, Basner M, Mason CE. "Molecular and physiological changes in the SpaceX Inspiration4 civilian crew." Nature. 2024 Aug 29;632(8027):1155-64. https://doi.org/10.1038/s41586-024-07648-x ; PMID: 38862026 ; PMCID: PMC11357997 , Aug-2024